

Algorithm 5: BIC-cherry + mode-aware expansion

Require: properly coloured digraph (G, σ) on L with the sicor-in-hub property,
mode $\in \{\text{strict}, \text{weak}\}$

Ensure: leaf-coloured network (N, σ)

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1:  $V(N) \leftarrow L \cup \{\rho\}$ ;  $E(N) \leftarrow \emptyset$ 
2: for all  $\{x, y\} \subseteq L$  with  $\sigma(x) \neq \sigma(y)$  do ▷ BIC-cherry network
3:   add  $p_{xy}$  with edges  $(\rho, p_{xy}), (p_{xy}, x), (p_{xy}, y)$ 
4: end for
5: for all  $\{x, y\} \subseteq L$  with  $\sigma(x) \neq \sigma(y)$  do ▷ expansion
6:    $(x', y') \leftarrow (\perp, \perp)$ 
7:   if mode = strict and  $(x, y), (y, x) \notin E(G)$  then  $(x', y') \leftarrow$   

   FINDJOINTCANDIDATES( $G, x, y$ )
8:   for all  $(a, b) \in ((x, y), (y, x))$  with  $(a, b) \notin E(G)$  do
9:     if  $b' = \perp$  then  $b' \leftarrow$  FINDCANDIDATE( $G, a, b$ , mode)
10:    add new vertex  $q$  with edges  $(p_{xy}, q), (q, a), (q, b')$  ▷  $[ab : ab']$ 
11:    remove edge  $(p_{xy}, a)$  ▷ shortcut
12:   end for
13: end for
14: return  $(N, \sigma)$ 

15: function FINDCANDIDATE( $G, a, b$ , mode)
16:    $C \leftarrow \emptyset$ 
17:   if mode = weak then  $C \leftarrow \{v \in L : \sigma(v) = \sigma(b), (a, v) \in E(G), (v, a) \in$   

 $E(G)\}$ 
18:   if  $C = \emptyset$  then  $C \leftarrow \{v \in L : \sigma(v) = \sigma(b), (a, v) \in E(G)\}$ 
19:   if  $C = \emptyset$  then  $C \leftarrow \{v \in L \setminus \{b\} : \sigma(v) = \sigma(b), (v, a) \in E(G)\}$ 
20:   if  $C = \emptyset$  then  $C \leftarrow \{v \in L \setminus \{b\} : \sigma(v) = \sigma(b)\}$ 
21:   return random element of  $C$ 
22: end function

23: function FINDJOINTCANDIDATES( $G, x, y$ )
24:    $X_1 \leftarrow \{v \in L : \sigma(v) = \sigma(x), (y, v) \in E(G)\}$ 
25:    $Y_1 \leftarrow \{v \in L : \sigma(v) = \sigma(y), (x, v) \in E(G)\}$ 
26:    $S \leftarrow \{(x', y') \in X_1 \times Y_1 : (x', y') \notin E(G), (y', x') \notin E(G)\}$ 
27:   if  $S \neq \emptyset$  then return random element of  $S$ 
28:    $X_2 \leftarrow \{v \in L \setminus \{x\} : \sigma(v) = \sigma(x)\}$ 
29:    $Y_2 \leftarrow \{v \in L \setminus \{y\} : \sigma(v) = \sigma(y)\}$ 
30:    $S \leftarrow \{(x', y') \in X_2 \times Y_2 : (x', y') \notin E(G), (y', x') \notin E(G)\}$ 
31:   if  $S \neq \emptyset$  then return random element of  $S$ 
32:   if  $X_1 \times Y_1 \neq \emptyset$  then return random element of  $X_1 \times Y_1$ 
33:   if  $X_2 \times Y_2 \neq \emptyset$  then return random element of  $X_2 \times Y_2$ 
34:   return  $(\perp, \perp)$ 
35: end function

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